Strategic Analysis of AMC 12A 2024 Problem 20

Using AMC12/COMC Strategic Problem Analysis Framework v2.1

Problem Statement

Points P and Q are chosen uniformly and independently at random on sides $\overline{AB}$ and $\overline{AC}$, respectively, of equilateral triangle $\triangle ABC$. Which of the following intervals contains the probability that the area of $\triangle APQ$ is less than half the area of $\triangle ABC$?

(A) $\left[\frac{3}{8}, \frac{1}{2}\right]$ **(B)** $\left(\frac{1}{2}, \frac{2}{3}\right]$ **(C)** $\left(\frac{2}{3}, \frac{3}{4}\right]$
(D) $\left(\frac{3}{4}, \frac{7}{8}\right]$ **(E)** $\left(\frac{7}{8}, 1\right]$

PROBLEM CONFIGURATION ANALYSIS

A. Problem Classification

- **Problem Type**: To Find (computational/estimation problem)
- **Competition Context**: AMC 12A, Problem 20
- **Difficulty Band**: Late-band (P17-25), specifically P20 - one of the hardest problems
- **Mathematical Domain**: Probability, Geometry, Integral Calculus (optional)
- **Estimated Time**: 6-8 minutes for experienced students; 10-12 minutes for those learning the technique

B. Problem Structure Identification

- **Given Information**:
  - Equilateral triangle $\triangle ABC$
  - Point P chosen uniformly at random on $\overline{AB}$
  - Point Q chosen uniformly at random on $\overline{AC}$
  - P and Q are independent random variables
- **Constraints**:
  - $P \in \overline{AB}$, $Q \in \overline{AC}$ (boundary constraints)
  - Uniform distribution for both points (geometric probability)
  - Independence of P and Q selections

- **Objective**: Find which interval contains the probability that $[\triangle APQ] < \frac{1}{2}[\triangle ABC]$
- **Hidden Assumptions**:
  - The answer asks for an interval, not exact value (estimation acceptable)
  - Uniform distribution on line segments corresponds to uniform distribution of distance parameters
  - Area ratio depends on product of two independent uniform random variables
- **Answer Choice Leverage**:
  - Intervals are well-separated, so rough estimation suffices
  - Can use visual/graphical reasoning to eliminate choices
  - Exact computation possible but not strictly necessary
  - Pattern: intervals partition $[3/8, 1]$ with increasing widths

C. Strategic Orientation

- **Hypothesis**: If we parameterize $AP = x$ and $AQ = y$ (with appropriate normalization), the condition becomes an inequality in the unit square
- **Conclusion**: The probability equals the area of a region in the unit square $[0, 1] \times [0, 1]$
- **Notation**:
  - Let side length of $\triangle ABC$ be 1 (WLOG normalization)
  - Let $AP = x$ where $x \in [0, 1]$
  - Let $AQ = y$ where $y \in [0, 1]$
  - $[\triangle ABC] = \frac{\sqrt{3}}{4}$ (area of unit equilateral triangle)
- **Intuitive Approach**:
  - Express area of $\triangle APQ$ in terms of x and y
  - Use the formula: Area $= \frac{1}{2} \cdot AP \cdot AQ \cdot \sin(\angle PAQ)$
  - Convert the area condition to an inequality: $xy \leq c$ for some constant c
  - Visualize as a region in the unit square
  - Estimate the area of this region
- **Cross-Domain Potential**:
  - Geometric Probability: Converting to area computation in sample space
  - Coordinate Geometry: Using (x, y) parameterization
  - Calculus: Computing area via integration (optional but more precise)
  - Estimation/Visualization: Graphical area approximation

KEY INSIGHT

The Core Insight: This is a geometric probability problem that transforms into computing the area under a hyperbola in the unit square. The key realization is that the area condition $[\triangle APQ] < \frac{1}{2}[\triangle ABC]$ becomes the simple inequality $xy < \frac{1}{2}$ in normalized coordinates.

STRATEGIC FRAMEWORK APPLICATION

A. Psychological Strategies

- **Mental Toughness**:
  - This is P20 - expect it to be challenging
  - Don't panic if the calculus approach seems difficult
  - Visual/geometric reasoning can provide the answer without integration
  - Trust that estimation is sufficient given the answer format
- **Creativity Cultivation**:
  - Multiple approaches available: exact integration vs. visual estimation
  - Can use symmetry and known geometric shapes to bound the area
  - Consider what the graph of $xy = \frac{1}{2}$ looks like in the unit square
  - Approximation by simpler curves (e.g., circular arc) can give bounds
- **Iterative Refinement**:
  - Start with rough sketch to eliminate obviously wrong answers
  - Refine estimate by comparing to known areas (half-square, quarter-circle)
  - If time permits, compute exactly via integration

B. Investigation Strategies

- **Startup Orientation**:
  - Read carefully: "uniformly and independently at random"
  - Identify that this is a 2D geometric probability problem
  - Recognize the need to parameterize positions of P and Q
- **Get Your Hands Dirty**:
  - Draw an equilateral triangle with points P and Q
  - Try specific positions: P at midpoint, Q at midpoint
  - Calculate area for this specific case
  - Ask: when is the area exactly half? This gives the boundary curve
- **Penultimate Step**:
  - The final step is comparing our computed/estimated probability to the given intervals

  - The penultimate step is determining the area of region $\{(x, y) : xy \leq \frac{1}{2}, x, y \in [0, 1]\}$

- **Make It Easier**:
  - Normalize the triangle to have side length 1
  - Use the fact that angles in equilateral triangle are known (60°)
  - Recognize that sine formula gives simple expression
- **Conjecture via Small Cases**:
  - When $P = A$ or $Q = A$: area is 0 (always less than half)
  - When $P = B$ and $Q = C$: area equals full triangle (not less than half)
  - When P at midpoint and Q at midpoint: $xy = \frac{1}{4}$, area $= \frac{1}{4}$ of original (less than half)
- **Visualization**:
  - Draw the unit square $[0, 1] \times [0, 1]$ representing (x, y) space
  - Sketch the curve $xy = \frac{1}{2}$, which is a hyperbola
  - The curve passes through $(0.5, 1)$ and $(1, 0.5)$
  - Shade the region where $xy \leq \frac{1}{2}$
  - Compare visually to half-square (0.5) and $\frac{3}{4}$ square

C. Argument Construction (Late-Band Playbook)

- **Late-Band Recognition**: This is P20, solidly in the late band
- **Pattern Matching**:
  - Geometric probability + uniform distribution → area computation
  - Triangle area with two sides and included angle → sine formula
  - Inequality constraint → region in parameter space
- **Transformation Chain**:
  - Physical setup → Parameterization (x, y)
  - Area condition → Algebraic inequality $(xy \leq \frac{1}{2})$
  - Probability → Geometric area in unit square
  - Exact or estimated area → Answer interval selection
- **Answer Choice Strategy**:
  - The intervals are mutually exclusive and cover different ranges
  - Visual inspection can eliminate (A), (B), (E)
  - Distinguish between (C) and (D) requires better estimate
  - Integration gives $\frac{\ln 2+1}{2} \approx 0.8466$ → answer (D)

WARNING

Common Pitfall #1: Forgetting the independence assumption. The probability is NOT the product of individual probabilities, but rather the area of a region in the joint sample space.
Common Pitfall #2: Incorrect area formula. Must use $\frac{1}{2}xy\sin(60^\circ) = \frac{\sqrt{3}}{4}xy$, not just $\frac{1}{2}xy$.
Common Pitfall #3: Computing exact integral when rough estimation suffices. This wastes time on a test.

TACTICAL METHODOLOGY SELECTION

A. Core Mathematical Tactics

- **Normalization Principle**:
  - Set side length to 1 (without loss of generality)
  - This simplifies all calculations
  - Ratios and probabilities are preserved under scaling
- **Symmetry Exploitation**:
  - The curve $xy = \frac{1}{2}$ has symmetry about the line $y = x$
  - This doesn't directly help but confirms the aesthetic of the problem
- **Extreme Case Analysis**:
  - When $x = 0$ or $y = 0$: always satisfied ($xy = 0 < \frac{1}{2}$)
  - When $x = 1, y = 1$: $xy = 1 > \frac{1}{2}$ (not satisfied)
  - This confirms we're looking at a proper subset of the unit square
- **Dimensional Analysis**:
  - Areas scale as length squared
  - Probability is dimensionless (ratio of areas)
  - This confirms our approach is dimensionally consistent
- **Invariance Recognition**:
  - The probability doesn't depend on which two sides we choose
  - Any two sides meeting at a vertex would give the same answer

B. Crossover Techniques

- **Geometry $\rightarrow$ Algebra**:
  - Geometric constraint (area inequality) becomes algebraic inequality ($xy \leq \frac{1}{2}$)
- **Probability $\rightarrow$ Geometry**:
  - Uniform probability on segments maps to uniform distribution on unit square

  - Probability computation reduces to area computation

- **Continuous → Discrete**:
  - While the problem involves continuous distributions, the answer is discrete (interval selection)

- **Exact → Approximate**:
  - Can solve exactly with calculus
  - Can estimate visually for interval identification
  - Strategy choice depends on time constraints

CONVERSION TECHNIQUES APPLICATION

A. Recognition Index

- **Primary Technique**: Geometric Probability via Area Computation
- **Secondary Technique**: Parametric Representation of Triangle Area
- **Advanced Technique**: Integration under a curve (optional)
- **Estimation Technique**: Visual approximation and bounds

B. Technique Selection

- **Step 1 - Parameterization**:
  - Represent P by parameter $x = \frac{AP}{AB}$ where $x \in [0, 1]$
  - Represent Q by parameter $y = \frac{AQ}{AC}$ where $y \in [0, 1]$
  - Sample space is the unit square $[0, 1] \times [0, 1]$

- **Step 2 - Area Formula**:
  - Use sine formula: $[\triangle APQ] = \frac{1}{2} \cdot AP \cdot AQ \cdot \sin(\angle A)$
  - For equilateral triangle: $\angle A = 60°$, $\sin(60°) = \frac{\sqrt{3}}{2}$
  - With normalization $AB = AC = 1$: $AP = x$, $AQ = y$
  - Therefore: $[\triangle APQ] = \frac{1}{2}xy \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}xy$

- **Step 3 - Inequality Conversion**:
  - Given: $[\triangle ABC] = \frac{\sqrt{3}}{4}$ (unit equilateral triangle)
  - Condition: $[\triangle APQ] < \frac{1}{2}[\triangle ABC]$
  - Substituting: $\frac{\sqrt{3}}{4}xy < \frac{1}{2} \cdot \frac{\sqrt{3}}{4}$
  - Simplifying: $xy < \frac{1}{2}$

- **Step 4 - Geometric Probability**:

- Probability $= \frac{\text{Area of favorable region}}{\text{Area of sample space}}$
- Sample space area $= 1 \times 1 = 1$
- Favorable region: $\{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1, xy \leq \frac{1}{2}\}$

- **Step 5 - Area Computation (Method A - Integration)**:
  - The boundary curve is $y = \frac{1}{2x}$
  - For $0 < x < \frac{1}{2}$: y ranges from 0 to 1 (curve above the square)
  - For $\frac{1}{2} \leq x \leq 1$: y ranges from 0 to $\frac{1}{2x}$ (curve inside square)
  - Area $= \int_0^{1/2} 1\, dx + \int_{1/2}^1 \frac{1}{2x}\, dx$
  - $= \frac{1}{2} + \frac{1}{2}[\ln x]_{1/2}^1 = \frac{1}{2} + \frac{1}{2}(\ln 1 - \ln \frac{1}{2})$
  - $= \frac{1}{2} + \frac{1}{2}(0 - (-\ln 2)) = \frac{1}{2} + \frac{\ln 2}{2} = \frac{1+\ln 2}{2}$
  - Numerical value: $\frac{1+0.693}{2} \approx 0.8466$

- **Step 5 - Area Computation (Method B - Visual Estimation)**:
  - Lower bound: The region includes the entire left half-square and bottom half-square minus overlap
  - This gives at least $\frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$
  - Upper bound: The region is less than the whole square (1)
  - The hyperbola $xy = \frac{1}{2}$ curves gently, bulging outward
  - Visual inspection suggests area is close to $\frac{7}{8}$ but slightly less
  - This places the answer in interval (D): $(\frac{3}{4}, \frac{7}{8}]$

C. Integration & Conflict Resolution

- **Consistency Check**:
  - Exact value $\frac{1+\ln 2}{2} \approx 0.8466$ confirms interval (D)
  - Visual estimate consistent with exact calculation
  - No conflicts between approaches

- **Trade-offs**:
  - Integration gives exact answer but requires calculus and time
  - Visual estimation is faster but requires good spatial intuition
  - On an exam, visual approach adequate for interval selection

PRO TIP

Time-Saving Strategy: On exam day, sketch the curve $y = \frac{1}{2x}$ in the unit square. Observe it passes through $(0.5, 1)$ and $(1, 0.5)$. The region below the curve is clearly more than $\frac{3}{4}$ of the square (eliminates A, B, C) but less than $\frac{7}{8}$ (eliminates E). Select (D) without computing

the integral. **Estimated time savings: 3-4 minutes.**

VERIFICATION STRATEGY DESIGN

A. Verification Level Selection

- **Chosen Level**: Level 4 (Sanity Check + Alternative Method)
- **Rationale**: This is P20 (late band, high stakes), and we need high confidence
- **Time Allocation**: 1-2 minutes

B. Verification Methods

Method 1: Boundary Check

- At $(0.5, 1)$: $xy = 0.5$, exactly on boundary ✓
- At $(1, 0.5)$: $xy = 0.5$, exactly on boundary ✓
- At $(1, 1)$: $xy = 1 > 0.5$, correctly excluded ✓
- At $(0, y)$ or $(x, 0)$: $xy = 0 < 0.5$, correctly included ✓

Method 2: Dimensional Analysis

- Probability must be dimensionless ✓
- Probability must be in $[0, 1]$ ✓
- Our answer $\approx 0.8466 \in [0, 1]$ ✓

Method 3: Limiting Cases

- If we required $[\triangle APQ] < \varepsilon$ (very small): probability $\to 0$ ✓
- If we required $[\triangle APQ] < [\triangle ABC]$: probability $\to 1$ ✓
- Our condition is intermediate, so probability between 0 and 1 is reasonable ✓

Method 4: Symmetry Verification

- The curve $xy = c$ is symmetric about $y = x$ ✓
- Our computation doesn't explicitly use this, but it's consistent
- If we chose P on AC and Q on AB instead, we'd get the same answer by symmetry ✓

Method 5: Answer Choice Elimination

- Visual inspection clearly shows area > 0.75 (eliminates A, B, C)
- Visual inspection suggests area < 0.875 (eliminates E)
- Only (D) remains
- Exact calculation confirms (D)

C. Confidence Calibration

- **Initial Confidence**: High (95%) - standard geometric probability problem
- **After Verification**: Very High (99%) - multiple methods agree
- **Final Decision**: Submit answer (D) with high confidence

SOLUTION ROADMAP

Step-by-Step Solution Plan

Phase 1: Setup and Parameterization (1-2 minutes)

1. Draw equilateral triangle $\triangle ABC$
2. Mark points P on $\overline{AB}$ and Q on $\overline{AC}$
3. Set side length to 1 (WLOG)
4. Define parameters: $x = AP$, $y = AQ$ where $x, y \in [0, 1]$
5. Recognize: uniform random selection $\Rightarrow$ uniform distribution on $[0, 1] \times [0, 1]$

Phase 2: Area Formula Derivation (1-2 minutes)

6. Apply sine area formula: $[\triangle APQ] = \frac{1}{2} \cdot AP \cdot AQ \cdot \sin(\angle PAQ)$
7. Note: $\angle PAQ = 60^\circ$ (angle of equilateral triangle)
8. Substitute: $[\triangle APQ] = \frac{1}{2} \cdot x \cdot y \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}xy$
9. Calculate: $[\triangle ABC] = \frac{\sqrt{3}}{4}$ (standard formula for unit equilateral triangle)

Phase 3: Inequality Conversion (1 minute)

10. Set up condition: $[\triangle APQ] < \frac{1}{2}[\triangle ABC]$
11. Substitute: $\frac{\sqrt{3}}{4}xy < \frac{1}{2} \cdot \frac{\sqrt{3}}{4}$
12. Simplify: $xy < \frac{1}{2}$
13. **Key insight**: The geometric condition reduces to a simple algebraic inequality

Phase 4: Probability Computation (2-4 minutes)
Option A: Visual Estimation (faster, sufficient for exam)

14. Sketch unit square with axes x and y
15. Draw hyperbola $xy = \frac{1}{2}$ passing through $(0.5, 1)$ and $(1, 0.5)$
16. Shade region where $xy \leq \frac{1}{2}$ (below/left of curve)
17. Observe: region includes entire left half and bottom half with overlap

18. Estimate visually: area appears to be approximately 0.85
19. Compare to intervals: clearly in $(\frac{3}{4}, \frac{7}{8}]$

Option B: Exact Integration (more rigorous)

20. Set up integral: Area $= \int_0^{1/2} 1\,dx + \int_{1/2}^1 \frac{1}{2x}\,dx$
21. Compute first integral: $\int_0^{1/2} 1\,dx = \frac{1}{2}$
22. Compute second integral: $\int_{1/2}^1 \frac{1}{2x}\,dx = \frac{1}{2}[\ln x]_{1/2}^1 = \frac{1}{2}(0 - \ln\frac{1}{2}) = \frac{\ln 2}{2}$
23. Sum: Total area $= \frac{1}{2} + \frac{\ln 2}{2} = \frac{1+\ln 2}{2}$
24. Evaluate: Using $\ln 2 \approx 0.693$, we get $\frac{1+0.693}{2} \approx 0.8466$
25. Identify interval: $0.75 < 0.8466 < 0.875$, so answer is (D)

Phase 5: Verification (1-2 minutes)

26. Check boundary points: $(0.5, 1)$ and $(1, 0.5)$ should satisfy $xy = 0.5$ ✓
27. Verify probability is in $[0, 1]$: $0.8466 \in [0, 1]$ ✓
28. Check reasonableness: Most of the unit square should satisfy $xy < 0.5$ ✓
29. Confirm answer choice: (D) $\left(\frac{3}{4}, \frac{7}{8}\right]$ ✓

Alternative Approaches

Alternative 1: Monte Carlo Simulation (if calculator allowed)

- Generate many random pairs (x, y) from $[0, 1] \times [0, 1]$
- Count proportion where $xy < 0.5$
- Should converge to approximately 0.846

Alternative 2: Complementary Probability

- Compute $P(xy > \frac{1}{2})$ instead
- This is the area above the hyperbola: $\int_0^{1/2} 0\,dx + \int_{1/2}^1 (1 - \frac{1}{2x})\,dx$
- $= \int_{1/2}^1 (1 - \frac{1}{2x})\,dx = [x - \frac{1}{2}\ln x]_{1/2}^1$
- $= (1 - 0) - (\frac{1}{2} - \frac{1}{2}\ln\frac{1}{2}) = 1 - \frac{1}{2} - \frac{\ln 2}{2} = \frac{1-\ln 2}{2}$
- Then $P(xy \le \frac{1}{2}) = 1 - \frac{1-\ln 2}{2} = \frac{1+\ln 2}{2}$ ✓

Common Pitfalls and How to Avoid Them

1. **Pitfall**: Using $[\triangle APQ] = \frac{1}{2}xy$ without the $\sin(60°)$ factor
   - **Consequence**: Wrong inequality $xy < \frac{\sqrt{3}}{2}$
   - **Prevention**: Always write out the sine formula explicitly
2. **Pitfall**: Forgetting to normalize the triangle side length
   - **Consequence**: Messy algebra with arbitrary constants
   - **Prevention**: State "WLOG, let side length = 1" at the beginning
3. **Pitfall**: Thinking the probability is $P(x < c) \cdot P(y < c')$ for some constants
   - **Consequence**: Wrong model of the problem
   - **Prevention**: Recognize this is about the joint distribution, not independent events
4. **Pitfall**: Computational error in the integral
   - **Consequence**: Wrong numerical value
   - **Prevention**: Double-check logarithm rules: $\ln(1/2) = -\ln 2$
5. **Pitfall**: Spending too much time on exact calculation
   - **Consequence**: Time pressure on remaining problems
   - **Prevention**: For P20, visual estimation is acceptable given interval format

COMPETITION STRATEGY INTEGRATION

A. Time Management Strategy

- **Target Time**: 6-8 minutes for this problem
- **Time Checkpoints**:
  - 2 min: Should have parameterization and area formula
  - 4 min: Should have inequality $xy < \frac{1}{2}$ and sketch drawn
  - 6 min: Should have answer selected (D) via estimation or integration
  - 8 min: Should complete verification and move on
- **Adjustment Rule**: If stuck after 10 minutes, mark answer (D) based on visual estimate and move on

B. Prioritization

- **Problem Positioning**: P20 is very late in the test
- **When to Attempt**:

  - Only after completing all problems you're confident about (P1-16)
  - Only if you have at least 10 minutes remaining
  - Skip if time is critically short
- **Expected Success Rate**:
  - Target audience: Top 5-10% of test-takers
  - Requires: Geometric probability + integration OR strong spatial reasoning

C. Risk Management

- **Risk Level**: Medium-High (P20, but tractable with right approach)
- **Mitigation Strategies**:
  - Use visual estimation instead of exact integration to save time
  - Answer choice spacing is generous - rough estimate suffices
  - Can eliminate obviously wrong intervals quickly
- **Confidence Threshold**: Proceed if confidence $\geq 60\%$ after 4 minutes of work

D. Abandonment Criteria

- **When to Abandon**:
  - If unable to derive $xy < \frac{1}{2}$ after 5 minutes
  - If cannot sketch or visualize the region after 6 minutes
  - If less than 8 minutes remain in the exam
- **Abandonment Strategy**:
  - Make educated guess based on partial work
  - Visual estimation suggests (D) is most likely even without full solution
  - Return to problem only if time permits after checking all other answers

E. Answer Recording

- **Format Check**: Multiple choice - bubble in (D)
- **Double-Check**: Verify you've selected an interval, not a point value
- **Final Confirmation**: $\left(\frac{3}{4}, \frac{7}{8}\right]$ contains approximately 0.8466

Complete Worked Solution

Approach 1: Visual Estimation (Exam Strategy - Fast)

Setup: Let the equilateral triangle have side length 1 (WLOG). Let P be at distance x from A along $\overline{AB}$, and Q be at distance y from A along $\overline{AC}$, where $x, y \in [0, 1]$.

Area Formula: Using the formula for the area of a triangle with two sides and included angle:

$$[\triangle APQ] = \frac{1}{2} \cdot x \cdot y \cdot \sin(60^\circ) = \frac{1}{2} \cdot x \cdot y \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4}xy$$

Since $[\triangle ABC] = \frac{\sqrt{3}}{4}$, the condition $[\triangle APQ] < \frac{1}{2}[\triangle ABC]$ becomes:

$$\frac{\sqrt{3}}{4}xy < \frac{1}{2} \cdot \frac{\sqrt{3}}{4}$$

$$xy < \frac{1}{2}$$

Probability as Area: The sample space is the unit square $[0, 1] \times [0, 1]$ with area 1. The favorable region is $\{(x, y) : xy \leq \frac{1}{2}\}$.

Visual Analysis: The boundary curve $xy = \frac{1}{2}$ is a hyperbola passing through $(0.5, 1)$ and $(1, 0.5)$. Sketching this curve:

- The region below/left of the curve clearly includes more than $\frac{3}{4}$ of the square
- But it's less than $\frac{7}{8}$ of the square (the hyperbola cuts off a noticeable corner)
- Visual estimation: approximately 0.84-0.85 of the square

Answer: The interval $\left(\frac{3}{4}, \frac{7}{8}\right]$ contains this probability. $\boxed{\textbf{(D)}}$

Approach 2: Exact Integration (Rigorous)

Setup: Same as Approach 1, leading to $xy < \frac{1}{2}$.

Integration Setup: The favorable region can be computed as:

$$\text{Area} = \int_0^1 \min\left(1, \frac{1}{2x}\right) dx$$

Split the integral at $x = \frac{1}{2}$ (where the curve intersects $y = 1$):

$$\begin{aligned}
\text{Area} &= \int_0^{1/2} 1\, dx + \int_{1/2}^1 \frac{1}{2x}\, dx \\
&= [x]_0^{1/2} + \frac{1}{2}\left[\ln x\right]_{1/2}^1 \\
&= \frac{1}{2} + \frac{1}{2}(\ln 1 - \ln \frac{1}{2}) \\
&= \frac{1}{2} + \frac{1}{2}(0 - (-\ln 2)) \\
&= \frac{1}{2} + \frac{\ln 2}{2} \\
&= \frac{1 + \ln 2}{2}
\end{aligned}$$

Numerical Evaluation: Using $\ln 2 \approx 0.693147$:

$$\frac{1+\ln 2}{2} \approx \frac{1+0.693}{2} \approx \frac{1.693}{2} \approx 0.8466$$

Interval Check:

$$\frac{3}{4} = 0.75 < 0.8466 < 0.875 = \frac{7}{8}$$

Therefore, the probability lies in the interval $\left(\frac{3}{4}, \frac{7}{8}\right]$.

Answer: $\boxed{\textbf{(D)} \ \left(\frac{3}{4}, \frac{7}{8}\right]}$

Approximation Note (for non-calculator exam)

If you don't have $\ln 2$ memorized, you can use:

$$\ln 2 = \ln(1+1) \approx 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{12-6+4-3}{12} = \frac{7}{12} \approx 0.583$$

Or more accurately (4 terms):

$$\ln 2 \approx 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} = \frac{60-30+20-15+12}{60} = \frac{47}{60} \approx 0.783$$

Wait, this series is for $\ln(1+x)$ with $x = 1$, which doesn't converge well. Better: use $\ln 2 \approx 0.69$ from memory or note that the answer must be in (D) from visual estimation alone.

Key Takeaways

1. **Geometric probability** often reduces to computing areas in parameter space
2. **Parameterization** is key: representing positions by normalized distances
3. **The sine formula** for triangle area is essential when two sides and included angle are known
4. **Visual estimation** can be sufficient for interval-based answer choices
5. **Integration** provides exact answers but takes more time
6. **Late-band problems** (P17-25) often require multiple steps and sophisticated reasoning
7. **Answer intervals** being well-separated means rough estimates are acceptable

Final Answer

The probability that the area of $\triangle APQ$ is less than half the area of $\triangle ABC$ is exactly $\frac{1+\ln 2}{2} \approx 0.8466$, which lies in the interval $\left(\frac{3}{4}, \frac{7}{8}\right]$.

$$\boxed{\textbf{Answer: (D)} \ \left(\frac{3}{4}, \frac{7}{8}\right]}$$